








Blockchain Applications

Lecture 6: Networks, Sentiment and Asset Classes



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Today's Question and Route

- **Question**: crypto assets do not move alone. How are they linked, are they an asset class of their own, and do protocol events and social media move prices?
- **Route**
  - networks: correlation trees, connectedness, Diebold–Yilmaz spillovers
  - are cryptos an asset class? factor analysis and classification
  - event studies: Ethereum upgrades
  - sentiment from social media: the BitMood index
- Data: End-of-Day Historical Data (EODHD) daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026; Ether upgrade dates 2019–2025

Learning Outcomes

- Build a correlation distance, a minimum spanning tree and an absorption ratio, and interpret hubs and connectedness
- Read a Diebold–Yilmaz spillover table: total, TO, FROM and NET spillovers
- Explain how factor analysis and classification separate cryptos from other assets
- Run and interpret a market-model event study: abnormal return (AR), cumulative abnormal return (CAR), individual and cross-sectional tests
- Describe how a text-sentiment index is built and why trading back-tests need out-of-sample selection

Abbreviations (1/2)

ACF autocorrelation function
ADA Cardano
AI artificial intelligence
AIC Akaike information criterion
ALGO Algorand
AR absorption ratio (networks)
AR abnormal return (event studies)
ATOM Cosmos
BH Benjamini–Hochberg (adjusted p-values)
Bi-LSTM bidirectional LSTM
BMP Boehmer–Musumeci–Poulsen (test)
BNB BNB (Binance)
BTC Bitcoin
CAPM capital asset pricing model
CAR cumulative abnormal return
CNN convolutional neural network
CS compound score (sentiment)
CS cross-sectional (in t_{CS})
CSAD cross-sectional absolute deviation
CTE conditional tail expectation
DAI Digital Art Index
DOGE Dogecoin
DOT Polkadot
EIP Ethereum Improvement Proposal
EODHD End-of-Day Historical Data (data provider)
ETC Ethereum Classic
ETF exchange-traded fund
ETH Ether (Ethereum)
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FIGARCH fractionally integrated GARCH
FIL Filecoin
FinBERT BERT language model for financial text
FTX FTX crypto exchange (collapsed November 2022)

Abbreviations (2/2)

FX foreign exchange
GARCH generalised autoregressive conditional heteroskedasticity
GRU gated recurrent unit
HAC heteroskedasticity and autocorrelation consistent
KMO Kaiser–Meyer–Olkin (sampling adequacy)
LDA latent Dirichlet allocation
LINK Chainlink
LSTM long short-term memory (network)
LTC Litecoin
MA moving average
MST minimum spanning tree
NFT non-fungible token
n.s. not significant
PC1, PC2 first, second principal component
RF random forest
RV realised volatility
RMSE root mean squared error
S&P 500 Standard & Poor's 500 stock index
SC synthetic control
SCAR standardised cumulative abnormal return
sd standard deviation
SE standard error
SHAP Shapley additive explanations
SI reaction-intensity score of a post (BitMood)
SOL Solana
SVM support vector machine
TRX Tron
USDT Tether (stablecoin)
VAR vector autoregression
VaR value at risk
vol volatility
WC engagement-weighted compound score (BitMood)
XMR Monero
XRP XRP Ledger token

Reading and Tools

- Asset classes: Pele, Wesselhöfft, Härdle, Kolossiatis & Yatracos (2023), *Are cryptos becoming alternative assets?*, European Journal of Finance
- Networks and spillovers: Mantegna (1999), Diebold & Yilmaz (2012); herding: Christie & Huang (1995), Chang, Cheng & Khorana (2000)
- Event studies: MacKinlay (1997); Lin, Găman, Wang & Pele (2026), *When does the market react to protocol innovation? Evidence from the Ethereum upgrade lifecycle*, under revision
- Sentiment: Conda, Găman, Bâg, Mazurencu-Marinescu-Pele, Pele & Härdle (2025), *BitMood: AI analysis of Bitcoin trends via Facebook emotions*, working paper
- Video units on Quantinar: Dynamic Crypto Networks, Cross-exchange Crypto Risk, Reddit Sentiment and BTC Returns, Elon Musk's Twitter Feed, DAI Digital Art Index
- Estimation: Python (`statsmodels` vector autoregression (VAR), `networkx`, `scikit-learn`); numbers marked "own" are estimated for this lecture

From Correlations to a Network

- Nodes: 19 coins; edges: dependence of daily log returns, January 2021 – October 2026 (EODHD, 2 121 days)
- Pearson correlation ρ_{ij} becomes a metric distance (Mantegna, 1999):

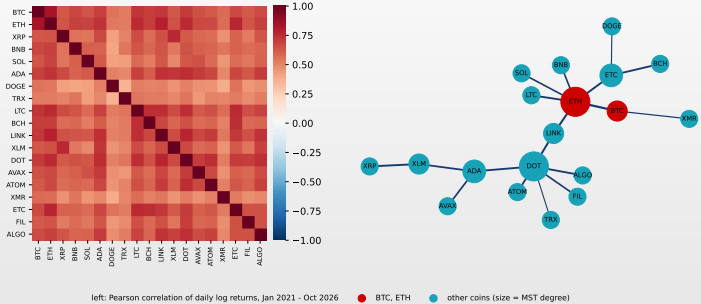
$$d_{ij} = \sqrt{2\,(1-\rho_{ij})}, \qquad \rho = 1 \Rightarrow d = 0, \quad \rho = 0 \Rightarrow d = \sqrt{2}$$

- Minimum spanning tree (MST): the $N-1$ edges that connect all coins with the smallest total distance

$$\mathrm{MST} = \arg \min_{T \text{ spanning tree}} \sum_{(i,j)\in T} d_{ij}$$

  - keeps the strongest links and removes the rest; a hub = a coin with many MST edges (high degree)
- Worked example: BTC–ETH, $\rho = 0.823$: $d = \sqrt{2 \cdot 0.177} = 0.595$; DOGE–TRX, $\rho = 0.349$: $d = 1.141$

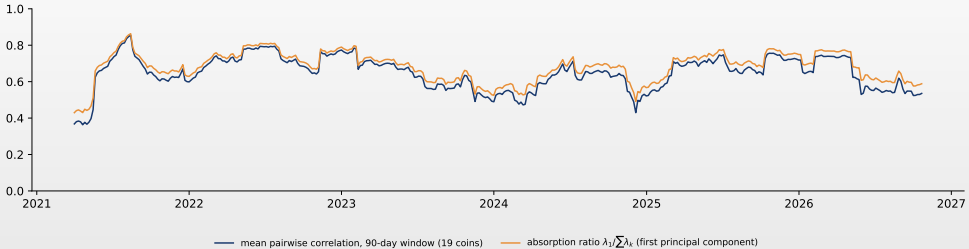
Correlation Matrix and Minimum Spanning Tree (own)



left: Pearson correlation of daily log returns, Jan 2021 - Oct 2026 BTC, ETH other coins (size = MST degree)

- Mean pairwise correlation 0.60 (minimum 0.35, DOGE–TRX); correlation with BTC from 0.52 (TRX) to 0.82 (ETH)
- Two hubs: ETH (degree 6: BTC, BNB, SOL, LTC, LINK, ETC) and DOT (degree 6: ADA, ATOM, FIL, ALGO, TRX, LINK); BTC links only to ETH and XMR

How Connected, and When? (own)



- Absorption ratio $\mathrm{AR}_t = \lambda_1 / \sum_k \lambda_k$ of the 90-day correlation matrix: the share of total variance explained by one common factor
- Mean correlation from 0.36 (April 2021) to 0.85 (mid-August 2021, after the May crash); 0.54 on 23 October 2026
- When one factor explains 80% of the variance, "diversifying across coins" buys almost nothing

Spillovers: Diebold–Yilmaz

- VAR(p) on N return series, $\mathrm{r}_t = \sum_{l=1}^{p} \Phi_l \mathrm{r}_{t-l} + \varepsilon_t$, moving-average (MA) form $\mathrm{r}_t = \sum_{h \geq 0} A_h \varepsilon_{t-h}$
- Generalised forecast-error variance decomposition (Pesaran–Shin), horizon H:

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathrm{e}_i' A_h \Sigma \mathrm{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathrm{e}_i' A_h \Sigma A_h' \mathrm{e}_i}, \qquad \tilde{\theta}_{ij} = \frac{\theta_{ij}}{\sum_j \theta_{ij}}$$

  - $\tilde{\theta}_{ij}$: share of coin i's forecast-error variance due to shocks in coin j; independent of variable ordering
- Indices (in %):

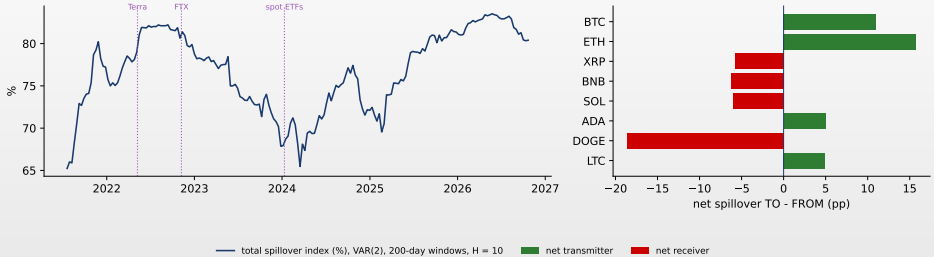
$$S = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}, \qquad \mathrm{TO}_j = \sum_{i \neq j} \tilde{\theta}_{ij}, \qquad \mathrm{FROM}_i = \sum_{j \neq i} \tilde{\theta}_{ij}, \qquad \mathrm{NET}_i = \mathrm{TO}_i - \mathrm{FROM}_i$$

Spillover Table, 8 Coins, 2021–2026 (own)

from ↓ / to →	BTC	ETH	XRP	BNB	SOL	ADA	DOGE	LTC	FROM
BTC	23.3	16.0	9.4	10.6	10.0	11.3	7.3	12.0	76.7
ETH	15.3	22.5	9.5	10.5	10.9	11.8	6.9	12.5	77.5
XRP	11.2	11.8	28.1	9.0	8.6	13.0	5.9	12.4	71.9
BNB	12.7	13.3	8.9	28.4	10.4	10.0	4.9	11.3	71.6
SOL	12.1	13.8	8.5	10.5	28.3	11.6	5.7	9.4	71.7
ADA	12.1	13.2	11.5	8.7	10.3	25.0	7.2	12.0	75.0
DOGE	11.4	11.1	7.3	6.0	7.1	10.4	36.4	10.2	63.6
LTC	12.8	14.0	11.0	9.9	8.3	11.9	7.1	25.0	75.0
TO	87.7	93.2	66.1	65.4	65.7	80.0	45.0	80.0	$S = 72.9$
NET	+11.0	+15.7	−5.8	−6.3	−6.0	+5.0	−18.6	+5.0	

- VAR(5) by Akaike information criterion (AIC), $H = 10$ days. Rows: whose variance is explained; columns: whose shocks explain it
- Worked check: $\mathrm{NET}_{ETH} = 93.2 - 77.5 = +15.7$; S = mean of the FROM column = 72.9%
- 73% of forecast-error variance comes from *other* coins: a highly connected system. ETH and BTC transmit; DOGE receives most

Spillovers through Time (own)

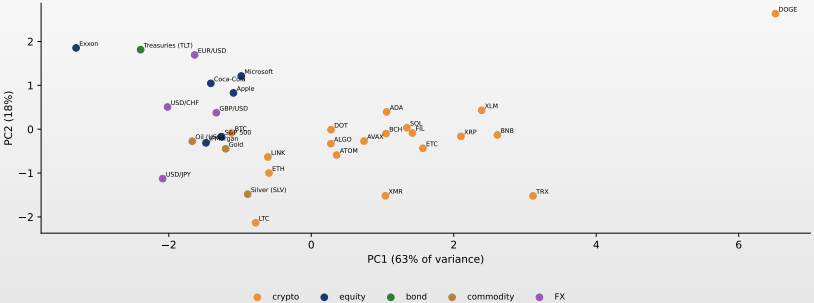


- Rolling VAR(2), 200-day windows: from 65.2% (July 2021) to 83.5% (May 2026); 80.4% on 23 October 2026
- Spillovers rose into Terra and FTX, fell during 2023, and climbed after the spot exchange-traded funds (ETFs): more common (institutional) flows, fewer coin-specific moves

Classifying Assets by Their Statistical "Phenotype" (own)

- Idea of Pele et al. (2023): describe each asset by the stylised facts of its returns, then ask whether cryptos form their own group
- Six features per asset, daily log returns 2021–2026: annualised volatility, skewness, log kurtosis, Hill tail index (top 5%), lag-1 autocorrelation (ACF) of $|r_t|$, 1% quantile in sd units
- 33 assets: 19 coins, 6 equities, Treasuries, gold, silver, oil, 4 foreign exchange (FX) rates; features standardised
- Principal components: first (PC1) explains 62.9%, second (PC2) 18.0%
  - PC1 loadings: vol 0.40, skew 0.39, kurtosis 0.45, tail index -0.43, ACF$|r|$ 0.46: a "wildness" factor
- k-means, $k = 3$:
  - cluster 1: all 14 traditional assets *and* BTC, ETH, LTC, LINK
  - cluster 2: 14 other coins; cluster 3: DOGE alone (skewness 5.9)

The Map of Assets (own)



- Bitcoin (vol 57%, tail index 3.56) sits next to the S&P 500 (16%, 3.44) on shape, not on scale: after standardisation, BTC's tails look like equity tails
- Most altcoins remain a separate cloud; DOGE is an outlier of its own

The Paper: Cryptos as a Separate Asset Class

- Pele, Wesselhöfft, Härdle, Kolossiatis & Yatracos (2023): 906 assets (234 cryptos, 635 stocks, 13 FX, 17 commodities, 5 bonds, 2 real-estate indices), daily, January 2014 – November 2020
- 24 indicators per asset, on expanding windows: variance, skewness, kurtosis, stable α and γ, left and right quantiles and conditional tail expectations (CTE) at 0.5–5%, $\rho(1)$, Hurst exponent, fractionally integrated GARCH (FIGARCH) d
- Factor analysis (principal components, Varimax; Kaiser–Meyer–Olkin (KMO) $= 0.92$): 3 factors explain 88%
  - **tail** factor 76% (quantiles, CTEs, variance, stable parameters); **memory** 6% (Hurst, $\rho(1)$, d); **moment** 6% (skewness, kurtosis)
- Logit $P(\text{crypto}) = \Lambda(\beta_0 + \beta_1 F)$: tail $\hat{\beta}_1 = 15.45$ (standard error (SE) 4.44), Nagelkerke $\tilde{R}^2 = 0.992$; moment -0.74 ($\tilde{R}^2 = 0.12$); memory n.s.
- Support vector machine (SVM) on two factors: 99.56% accuracy, only BTC and USDT misclassified; k-means (6 clusters): 98.45%
- Conclusion: a separate asset class, "mainly due to the tail factor"; BTC lies at the border

What Separates Cryptos? Averages by Class (Paper)

	Cryptos	Stocks	Bonds	FX	Commodities	Real estate
$\sigma^2 \cdot 10^3$	26.442	0.400	0.745	0.028	0.388	0.146
stable α	1.342	1.602	1.460	1.722	1.662	1.696
$Q_{1\%}$	−0.366	−0.054	−0.057	−0.013	−0.050	−0.034
$\mathrm{CTE}_{1\%}$	−0.564	−0.085	−0.097	−0.017	−0.074	−0.052
skewness	0.686	−0.560	−0.829	−0.937	−0.495	−1.142
FIGARCH d	0.553	0.289	0.545	0.407	0.426	0.510

- Crypto variance is about 60 times larger; stable $\alpha = 1.34$: much heavier tails (Lecture 2: smaller tail index)
- Separation by projection pursuit: without stable α no direction separates cryptos perfectly; with it, 181 618 directions do
- Average 1% value at risk (VaR) of an equal-weight portfolio (historical): classical 3.01%, cryptos 16.34%, mixed 4.37%
- Own update (previous slides): standardising by volatility puts BTC and ETH next to traditional assets in 2021–2026, consistent with "BTC at the border"; most altcoins stay apart

The Market-Model Event Study

- Normal return from a market model estimated on $[-250, -31]$ days before the event:

$$R_t = \alpha + \beta R_{m,t} + u_t, \qquad \mathrm{AR}_t = R_t - \hat{\alpha} - \hat{\beta} R_{m,t}, \qquad \mathrm{CAR}(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} \mathrm{AR}_t$$

- One event: $t = \mathrm{CAR}(\tau_1, \tau_2) / \sqrt{L\, \hat{\sigma}_u^2}$, $L = \tau_2 - \tau_1 + 1$
- Many events $e = 1, \ldots, N$: cross-sectional (CS) test and standardised Boehmer–Musumeci–Poulsen (BMP) test on standardised CARs (SCAR)

$$t_{CS} = \frac{\overline{\mathrm{CAR}}}{s_{\mathrm{CAR}}/\sqrt{N}}, \qquad \mathrm{SCAR}_e = \frac{\mathrm{CAR}_e}{\hat{\sigma}_{u,e}\sqrt{L}}, \qquad t_{BMP} = \frac{\overline{\mathrm{SCAR}}}{s_{\mathrm{SCAR}}/\sqrt{N}}$$

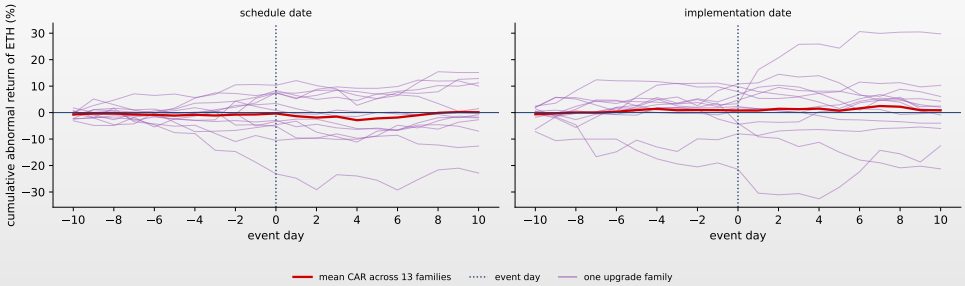
- Sign test: number of positive CARs $\sim \mathrm{Bin}(N, 0.5)$ under H_0
- Own design: Ether against an equal-weighted index of 11 other coins, daily; 13 upgrade families (Istanbul 2019 to Fusaka 2025), two dates each: formal schedule and implementation

CARs around 13 Upgrades (own)

implementation	CAR$[-1,1]$	t
Istanbul (Dec 2019)	0.4%	0.12
Muir Glacier	−0.3%	−0.10
Beacon Chain	1.3%	0.41
Berlin	−9.9%	−1.79
London (EIP-1559)	8.7%	1.35
Altair	4.4%	0.84
Arrow Glacier	−4.3%	−0.85
Gray Glacier	−4.6%	−1.48
Merge (Sep 2022)	−11.1%	−3.09
Shapella	3.3%	1.17
Dencun	−3.1%	−1.16
Pectra (May 2025)	11.3%	2.92
Fusaka (Dec 2025)	1.7%	0.54

- Cross-section, implementation:
  - $\overline{\mathrm{CAR}}[-1,1] = -0.18\%$, $t_{CS} = -0.10$ (p 0.92), $t_{BMP} = -0.20$
  - 7 of 13 positive (sign test p = 1.00)
  - $\overline{\mathrm{CAR}}[0,5] = -0.21\%$ (p 0.94)
- Schedule dates: $\overline{\mathrm{CAR}}[-1,1] = -0.59\%$ (p 0.63)
- Implementation minus schedule, paired: +0.41% (p 0.84)
- Two large, opposite reactions (Merge, Pectra) cancel: no common revaluation

Abnormal Returns around the Event Day (own)



- Individual paths fan out in both directions; the mean path stays near zero
- Estimated betas of Ether to the coin index: 0.75 (Dencun) to 1.20 (Fusaka); residual sd 1.5–3.7% a day

The Paper: Attention, Activity, Volatility, Not Price

- Lin, Găman, Wang & Pele (2026): 13 upgrade families; outcomes beyond returns; cross-asset difference-in-differences and synthetic control (SC) against donor coins

$$\hat{\delta}_{e,s} = \Delta y_{e,s,ETH} - \frac{1}{J_e}\sum_{j=1}^{J_e} \Delta y_{e,s,j}, \qquad \hat{y}^{SC}_{e,t} = \sum_j w_{e,j}\, y_{j,t},\ w_{e,j} \geq 0,\ \textstyle\sum_j w_{e,j} = 1$$

- Exact sign-flip tests at family level, Benjamini–Hochberg adjusted p-values (BH p)

outcome (log points)	estimate	BH p	meaning
Google attention, schedule day	0.184	0.034	search starts at scheduling
Google attention, implementation day	0.350	0.004	+42%
active addresses, implementation (synthetic control)	−0.145	0.020	−13.5%: users pause
median hourly log realised volatility (RV), implementation hours	0.316	0.007	+37% volatility
jump share, bipower variation (24 h)	≈ 0	0.99	no lasting change
paired CAR, implementation minus schedule	−0.0098	raw p 0.375	no repricing

- "The absence of CAR is economically informative": anticipated upgrades are priced; the event moves attention, activity and volatility

BitMood: Building a Sentiment Index

- Conda, Găman, Bâg, Mazurencu-Marinescu-Pele, Pele & Härdle (2025): 120 000 English Facebook posts on Bitcoin (5% monthly sample of about 2.3 million, CrowdTangle), January 2015 – October 2023
- FinBERT gives each post probabilities S^+, S^0, S^-; compound score (CS) $\mathrm{CS}_i = S_i^+ - S_i^-$
- Engagement-weighted compound score (WC), daily, with T_i = total interactions of post i:

$$P_d^{\pm,0} = \frac{\sum_{i \in d} S_i^{\pm,0} T_i}{\sum_{i \in d} T_i}, \qquad \mathrm{WC}_d = \frac{P_d^+ - P_d^-}{P_d^+ + P_d^- + P_d^0} \in [-1, 1]$$

- Reaction-intensity score (SI) of a post (weights from the literature): $\mathrm{SI}_i = \sum_k (\log N_{i,k} + 1)\, w_k$; like 0.1, comment 0.05, share 0.2, love/wow/haha 0.5, sad/angry -0.5
- Topics (latent Dirichlet allocation, LDA): investment 42.8%, trading and blockchain 24.6%, gaming 14.3%, social-media links 13%, business 5.3% of tokens

BitMood: What Sentiment Can and Cannot Do

- Contrarian rule (after Preis et al.): $\Delta f(t, \delta t) = f(t) - \frac{1}{\delta t} \sum_{k=1}^{\delta t} f(t-k)$; buy when sentiment fell, sell when it rose
- Best configuration in the 2023 test period: $\delta t = 3$ days, compound sentiment, cost 10^{-5}: cumulative return 1.321, Sharpe 0.210
- Volume forecasts (root mean squared error, RMSE, scaled volume; revised version) of linear regression, random forest (RF), long short-term memory (LSTM), convolutional (CNN) and gated recurrent unit (GRU) networks:

	Lin. reg.	RF	stacked LSTM	CNN	LSTM-CNN	GRU	Bi-LSTM
validation	0.1423	0.1767	0.2323	0.2179	0.2501	0.1840	0.1701
test	0.1935	0.2355	0.2453	0.2578	0.2766	0.1959	0.1930

- The linear regression ties the best deep network (difference 0.0005): "model complexity does not guarantee superior prediction"
- Caveat for any back-test: the best of many window–feature–cost combinations (7 windows $\times$ 6 costs $\times$ several sentiment series) was chosen on the same 2023 period it is scored on; an honest test selects on earlier data (Lecture 4: threshold chosen on validation)
- Sentiment helps forecast *volume* (lagged volume and negative sentiment dominate the Shapley additive explanations, SHAP); there is no evidence here on returns

Summary

- Crypto returns form a dense network: mean correlation 0.60; ETH and DOT are the hubs of the minimum spanning tree; one factor explains 68% of the variance on average (59% in October 2026)
- Spillovers: 73% of forecast-error variance comes from other coins; ETH and BTC transmit, DOGE receives; the index rose to 83.5% in 2026
- Cryptos are a separate asset class through their tails (paper: tail factor 76%, SVM 99.6%); after scaling, BTC and ETH move towards classical assets
- Ethereum upgrades: large CARs of both signs, mean zero; the paper finds the effect in attention, activity and volatility instead
- Sentiment helps forecast volume; trading claims need out-of-sample selection

Interpretation Exercises

1. The MST shows BTC with only two links while ETH has six. Does this mean BTC is less important for the market? Use the spillover table in your answer.
2. The absorption ratio rises from 0.45 to 0.85. What happens to the benefit of holding ten coins instead of one?
3. In the spillover table, $\mathrm{NET}_{DOGE} = -18.6$. Explain what this number measures and what it does not prove.
4. The Merge has CAR$[-1, 1] = -11.1\%$ ($t = -3.09$) and Pectra $+11.3\%$ ($t = 2.92$), while the mean over 13 upgrades is -0.18% (p 0.92). What can and cannot be concluded?
5. A sentiment strategy reports a Sharpe ratio of 0.21 in 2023. Which two questions would you ask before using it?

⊡ Next: Lecture 7, machine learning, artificial intelligence (AI) and their limits; exam review

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Do Investors Follow the Crowd? The CSAD Test
- Herding Is Not a Constant (own)
- Why Not Just the Median Price?
- Worked Example: Composition Bias in a Boom (simulated)

Do Investors Follow the Crowd? The CSAD Test

- Cross-sectional absolute deviation of returns around the market return $R_{m,t}$:

$$\mathrm{CSAD}_t = \frac{1}{N}\sum_{i=1}^{N} |R_{i,t} - R_{m,t}|$$

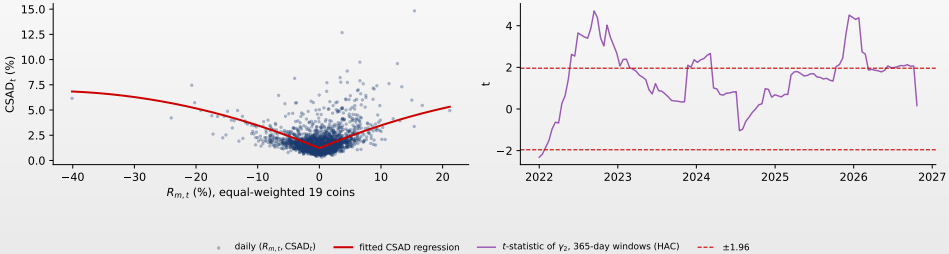
- Rational pricing (capital asset pricing model, CAPM): dispersion rises linearly with $|R_{m,t}|$. Herding: in big moves, coins move together and dispersion rises *less*

$$\mathrm{CSAD}_t = \alpha + \gamma_1 |R_{m,t}| + \gamma_2 R_{m,t}^2 + \varepsilon_t, \qquad H_0 : \gamma_2 = 0, \quad \text{herding: } \gamma_2 < 0$$

- Own estimates, 19 coins, equal-weighted R_m, $n = 2\,121$, heteroskedasticity and autocorrelation consistent (HAC(10)) errors:

sample	$\hat{\gamma}_1$ (t)	$\hat{\gamma}_2$ (t)	R^2
all days	0.254 (10.76)	−0.0028 (−2.88)	0.248
up days ($R_m > 0$)		0.0003 (0.03)	
down days		−0.0001 (−0.12)	

Herding Is Not a Constant (own)



- Full sample: $\hat{\gamma}_2 < 0$ significant; the fitted curve is concave (vertex at $|R_m| = \hat{\gamma}_1/2|\hat{\gamma}_2| \approx 45\%$), and the curvature rests on the 16 days with $|R_m| > 15\%$
- 365-day windows: γ_2 significantly negative in only 1.7% of windows, significantly *positive* (anti-herding) in 39%
- Interpretation: herding in crypto is an episode, not a trait; a full-sample coefficient can hide the opposite behaviour most of the time

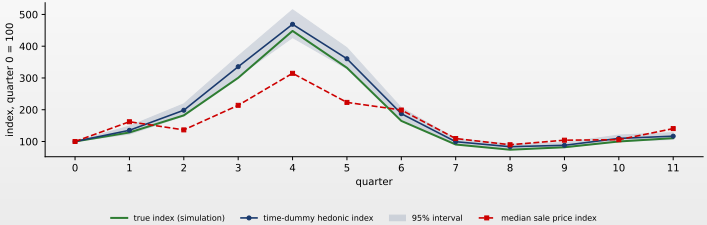
Why Not Just the Median Price?

- Each non-fungible token (NFT) or digital artwork is unique and trades rarely; the works sold this quarter differ from those sold last quarter
- Median sale price mixes two things: price changes and *composition* changes
- Time-dummy hedonic regression on all sales n in quarters $t = 0, \ldots, T$:

$$\ln P_n = \sum_{t=1}^{T} \delta_t D_{n,t} + x_n' \beta + \varepsilon_n, \qquad I_t = 100\, e^{\hat{\delta}_t}$$

  - x_n: characteristics (rarity, artist, size, collection); $\hat{\delta}_t$: quality-adjusted log price level
- Used for the Digital Art Index (Quantinar unit) and for real estate and art markets in general

Worked Example: Composition Bias in a Boom (simulated)



- 3 600 simulated sales, 12 quarters; in the boom (quarters 2–5) more low-rarity, unknown-artist works are sold
- Hedonic regression recovers the characteristics (rarity 2.52, known artist 1.20; true 2.5 and 1.2) and the index: RMSE 0.09 in logs
- Median index: RMSE 0.25; at the peak it shows 315 instead of 448 (30% too low): the boom looks smaller because cheaper works flood the market